One Pigeonhole for 3-SAT: A Determinant That Removes the Correction

The Proof

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OpenAI and Claude (Anthropic) first announced this theorem. This note gives a short proof. The ideas behind it, the comparison with earlier work, a worked example, and limits are in the full paper. The Lean 4 library `OnePigeonhole` in the same repository formalizes this proof down to a multitape machine, in a network form, with other parameters, with a branching tree in place of the citation of the sparsification lemma, and with a table in place of determinant code.

Theorem 1. *For every $\varepsilon > 0$, 3-SAT on n variables is decided deterministically in time $C_\varepsilon |\varphi|^{O_\varepsilon(1)} 2^{\varepsilon n}$, where $|\varphi|$ is the size of the input φ.*

1 The formula

Fix $\Delta \geq 3$ and an integer $m \geq \max(60, \Delta)$, and put $Q = m^4$, $R = m^3$, $L = 6Q$, $K = L^Q$. Take a formula with at most three literals per clause, n variables and at most Δ occurrences per variable, so with $M \leq \Delta n$ occurrences. Put $r = \Delta n/m$. We count its solutions. If $n < K^2$, enumerate.

Each occurrence j has a bit x_j, which is 1 if the literal of j is false, and a free bit y_j. Then

$$\#\text{solutions} = \sum_{\text{variables}} \sum_{y} \prod_{e} \Big(1 - \prod_{j \in e} y_j\Big) \prod_{j} \mathbf{1}[x_j = y_j]. \tag{1}$$

Words. Words are $[L]^Q$. Give the variables words $s(v)$ cyclically. Since $n \geq K^2$, a word has at most $2n/K$ variables and $2\Delta n/K$ occurrences. Let the *load* of (u, w) be the number of occurrences placed so far of variables with the word u in clauses with the word w. For fixed u, the loads sum over w to at most $2\Delta n/K$. Going through the clauses, give the clause e a word $o(e)$ that differs in every digit from the words of its variables; at least $(L-3)^Q \geq K(1 - 3Q/L) = K/2$ words do. Among these, take the word w with the least sum of $\text{load}(s(v), w)$ over the variables v of e. This sum has average at most $3(2\Delta n/K)/(K/2) = 12\Delta n/K^2$, and e adds at most $3 \leq \Delta n/K^2$ to a load. So all loads are at most $h = \lceil 13\Delta n/K^2 \rceil$.

Cells. A *cell* is (u, w, λ) with words u, w that differ in every digit and $\lambda \in [h]$. There are $N \leq K^2 h \leq 14\Delta n$ cells. Since loads are at most h, each occurrence j of v in e gets its own cell $c_j = (s(v), o(e), \lambda)$. Put $d_j = (o(e), s(v), \lambda)$.

2 The window

For $E \subseteq [Q]$, let τ_E exchange the letters of the two words of a cell in the digits of E. So $\tau_{[Q]} c_j = d_j$ and $\tau_E \tau_F = \tau_{E+F}$, with $E + F$ the symmetric difference. Since the two words differ in every digit, the cells $\tau_E c$ for different E are different.

Lemma 1. *There are rationals κ_E, $|E| \leq R$, and functions α_l, f_l on cells, with at most N/m^2 indices l, such that for all cells c, d*

$$\sum_{|E| \leq R} \kappa_E \mathbf{1}[\tau_E c = d] - \mathbf{1}[\tau_{[Q]} c = d] = \sum_{l} \alpha_l(c) f_l(d).$$

Proof. Put $b = 2^Q$. For $\sigma \in \{\pm 1\}^Q$ put $\sigma^E = \prod_{t \in E} \sigma_t$. Then $\sigma^E \sigma^F = \sigma^{E+F}$, and $\sum_\sigma \sigma^E$ is b if $E = \varnothing$ and 0 otherwise. If σ has k minus signs, then $\sigma^{[Q]} = (-1)^k$ and $\sum_t \sigma_t = Q - 2k$. Let p be a polynomial of degree at most R with $p(k) = (-1)^k$ on the at most $R+1$ integers k with $|Q - 2k| \leq R$. Put $k = \sum_t (1 - \sigma_t)/2$ in p, expand, and use $\sigma_t^2 = 1$. This gives κ_E such that $\rho_\sigma := \sum_{|E| \leq R} \kappa_E \sigma^E - \sigma^{[Q]}$ is 0 unless $|\sum_t \sigma_t| > R$. Call such σ *exceptional*. Since $\sum_\sigma (\sum_t \sigma_t)^2 = bQ$, at most $bQ/R^2 = b/m^2$ vectors σ are exceptional.

The *cube* of a cell c_0 is the set of the b cells $\tau_F c_0$. The cubes partition the cells, so there are N/b cubes. Choose a base cell c_0 in each cube. If $c = \tau_F c_0$ and $d = \tau_{F'} c_0$, then for every E we have

$\mathbf{1}[\tau_E c = d] = \mathbf{1}[E + F + F' = \varnothing] = b^{-1}\sum_\sigma \sigma^E \sigma^F \sigma^{F'}$. So the left side of the lemma is $b^{-1}\sum_\sigma \rho_\sigma \sigma^F \sigma^{F'}$, a sum over exceptional σ. If c, d lie in different cubes, the left side is 0. So take as indices the pairs l of a cube and an exceptional σ, with $\alpha_l(\tau_F c_0) = \rho_\sigma \sigma^F / b$ and $f_l(\tau_{F'} c_0) = \sigma^{F'}$ on this cube, and both functions 0 outside it. There are at most $(N/b)(b/m^2) = N/m^2$ indices. □

3 Trading and the pigeonhole

Fix values of all bits. Let A, D be the $M \times M$ matrices with

$$A_{ij} = \sum_{|E| \le R} \kappa_E \mathbf{1}[\tau_E c_i = d_j]\, \mathbf{1}[x_i = y_j], \qquad D_{ij} = \mathbf{1}[\tau_{[Q]} c_i = d_j]\, \mathbf{1}[x_i = y_j].$$

Since $\tau_{[Q]} c_i = d_i$ and the cells d_j are different, D is diagonal and $\det D = \prod_j \mathbf{1}[x_j = y_j]$. By Lemma 1, $(A - D)_{ij} = \sum_l \alpha_l(c_i) f_l(d_j) \mathbf{1}[x_i = y_j]$. So every row of $A - D$ lies in the span of the at most $2N/m^2$ vectors $(f_l(d_j)\mathbf{1}[\beta = y_j])_j$, $\beta \in \{0, 1\}$, and $2N/m^2 \le 28\Delta n/m^2 \le r$.

Expand $\det D = \det(A - (A - D))$ by linearity in the rows. For a set T of rows, let $\langle T \rangle$ have the rows in T from $A - D$ and the other rows from A. More than r rows of $A - D$ are dependent, so

$$\prod_j \mathbf{1}[x_j = y_j] = \sum_{|T| \le r} (-1)^{|T|} \det\langle T \rangle. \tag{2}$$

Fix T. An entry (i, j) of $\langle T \rangle$ can be nonzero only if $\tau_E c_i = d_j$ for some E; this E is unique, and we call it E_{ij}. In a row outside T, a nonzero entry has $|E_{ij}| \le R$. In a row in T, it has $|E_{ij}| \le R$ or $E_{ij} = [Q]$. Let $z_1, \ldots, z_Q$ be formal variables, and let $\langle T \rangle_z$ be $\langle T \rangle$ with each entry (i, j) multiplied by $\prod_{t \in E_{ij}} z_t$. In the expansion of $\det\langle T \rangle_z$ over permutations, a permutation π gives a monomial z^a in which a_t is the number of rows i with $t \in E_{i\pi(i)}$. So

$$\sum_t a_t = \sum_i |E_{i\pi(i)}| \le MR + |T|Q \le \Delta n R + rQ = 2rQ,$$

since $R = Q/m$, and some $a_t \le 2r$. For a polynomial f in z, let $\mathrm{Sum}_t(f)$ be the sum of its coefficients at the monomials z^a with $a_t \le 2r$, and $a_{t'} > 2r$ for $t' < t$. Each monomial of $\det\langle T \rangle_z$ is counted for exactly one t, and the sum of all its coefficients is $\det\langle T \rangle$. So $\det\langle T \rangle = \sum_t \mathrm{Sum}_t(\det\langle T \rangle_z)$.

Fix t. For a set S of rows, let $H_{T,t,S}$ be $\langle T \rangle_z$ with only these entries kept: in the rows of S those with $t \in E_{ij}$, and in the other rows those with $t \notin E_{ij}$. By linearity in the rows, $\det\langle T \rangle_z = \sum_S \det H_{T,t,S}$. Each monomial of $\det H_{T,t,S}$ has $a_t = |S|$, so $\mathrm{Sum}_t(\det H_{T,t,S}) = 0$ for $|S| > 2r$. Hence

$$\prod_j \mathbf{1}[x_j = y_j] = \sum_{|T| \le r} (-1)^{|T|} \sum_t \sum_{|S| \le 2r} \mathrm{Sum}_t\big(\det H_{T,t,S}\big). \tag{3}$$

4 The determinant across a digit

Equation (3) holds for all values of the bits. Now let the bits vary. Fix T, t, S and $H = H_{T,t,S}$. Then $H_{ij} = \gamma_{ij} \mathbf{1}[x_i = y_j]$, where γ_{ij} is zero or a rational times a monomial in z, and does not depend on the bits. For an occurrence i of v in e, let s_i, o_i be the letters of $s(v), o(e)$ at digit t. If $\gamma_{ij} \ne 0$, comparing first words at digit t in $\tau_{E_{ij}} c_i = d_j$ gives $o_j = s_i$ if $t \notin E_{ij}$, and $o_j = o_i$ if $t \in E_{ij}$; and $t \in E_{ij}$ exactly when $i \in S$. So with $g_i = s_i$ for $i \notin S$, and $g_i = o_i$ for $i \in S$,

$$\gamma_{ij} \ne 0 \quad \Longrightarrow \quad o_j = g_i. \tag{4}$$

The *bits of a letter* g are the variables v with $s(v)_t = g$ and the bits y_j with $o_j = g$. Each bit belongs to exactly one letter. For $\xi \in \{0, 1\}^S$, let H^ξ be H with x_i replaced by the number ξ_i for $i \in S$. Then $\det H = \sum_\xi \det H^\xi \prod_{i \in S} \mathbf{1}[x_i = \xi_i]$.

Let H_g^ξ be the submatrix of H^ξ with the rows that have $g_i = g$ and the columns that have $o_j = g$. By (4), all entries of H^ξ outside these submatrices are 0. An entry of H_g^ξ reads y_j with $o_j = g$ and, if $i \notin S$, reads x_i with $s_i = g$, so it reads only bits of g. Call S *balanced* if every H_g^ξ is square; this depends only on t, S. There are M rows and M columns, so if S is not balanced, some H_g^ξ has more rows than columns. These rows are dependent, and $\det H^\xi = 0$. If S is balanced, listing the rows and the columns by letter makes H^ξ block diagonal, so $\det H^\xi = \operatorname{sgn}(t,S) \prod_g \det H_g^\xi$, with a sign that depends only on t, S. Put

$$\Psi_g = \prod_{i \in S:\, s_i = g} \mathbf{1}[x_i = \xi_i] \prod_{e:\, o(e)_t = g} \Big(1 - \prod_{j \in e} y_j\Big) \det H_g^\xi,$$

a polynomial in z whose coefficients depend only on the bits of g. Over all g, each indicator and each clause factor appears exactly once. So the sum of $\prod_g \Psi_g$ over all bits equals $\prod_g \sum_{\text{bits of } g} \Psi_g$. With (1), (3) and the linearity of Sum_t, this gives

$$\#\text{solutions} = \sum_{|T| \le r} (-1)^{|T|} \sum_t \sum_{\substack{|S| \le 2r \\ S \text{ balanced}}} \operatorname{sgn}(t,S) \sum_{\xi \in \{0,1\}^S} \operatorname{Sum}_t\Big(\prod_g \sum_{\text{bits of } g} \Psi_g\Big). \tag{5}$$

5 The cost and the end of the proof

We evaluate (5). Since $M/m \le r$, for every $q > 0$ we have $\sum_{k \le qr} \binom{M}{k} \le m^{qr} \sum_k \binom{M}{k} m^{-k} = m^{qr}(1+1/m)^M \le m^{qr} \mathrm{e}^r$. So there are at most $(\mathrm{e}m)^r$ sets T and at most $4^r(\mathrm{e}m^2)^r$ pairs (S, ξ), so at most $(4\mathrm{e}^2 m^3)^r$ triples (T, S, ξ).

A letter g is the letter at digit t of K/L words, so it has at most $2n/L$ variables. At most $K^2 h/L \le 14\Delta n/L$ cells have the letter g at digit t of the first word, and the cells d_j are different, so g has at most $14\Delta n/L$ bits y_j. So g has at most $2^{16\Delta n/L} \le 2^{n/m}$ assignments, since $L = 6m^4$ and $\Delta \le m$.

The numbers κ_E depend only on m. All polynomials have exponents at most M in each z_t, so they have at most $(M+1)^Q$ monomials, with coefficients of polynomially many bits, and a determinant of such polynomials is found from its values at $(M+1)^Q$ points. So one Ψ_g, the product over the L letters, one Sum_t, and the words and signs each take time polynomial in n, with degree depending on m, and this is at most $C_m 2^{n/m}$. Multiply by the triples, the Q digits, the L letters and their assignments. Since $2n/m \le r$ and $8\mathrm{e}^2 < 60 \le m$, one formula takes time at most $C_m(8\mathrm{e}^2 m^3)^r \le C_m m^{4r} = C_m 2^{4\Delta n \log_2 m/m}$ times a polynomial.

Fix $\varepsilon > 0$. By the sparsification lemma of Impagliazzo, Paturi and Zane (*J. Comput. System Sci.* 63 (2001), p. 513), there is $\Delta \ge 3$ such that a 3-CNF on n variables is satisfiable if and only if one of at most $2^{\varepsilon n/2}$ 3-CNFs is, each with at most Δ occurrences per variable, computed in time $2^{\varepsilon n/2}$ times a polynomial. Choose $m \ge \max(60, \Delta)$ with $4\Delta \log_2 m/m \le \varepsilon/2$. For each of these formulas, run the algorithm above. The 3-CNF is satisfiable if and only if one of them has a solution, and the total time is at most $C_\varepsilon |\varphi|^{O_\varepsilon(1)} 2^{\varepsilon n}$. □